NOETHER NORMALIZATION OVER THE FIBRE: FACTORIZATION THROUGH AFFINE SPACE WITH FINITE FIBRE MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the construction of the factorization; no step is referred to the reader.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-noether-normalization-affine is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Noether normalization for a finite type algebra over a field. Algebra, Lemma algebra-lemma-Noether-normalization, invoked at proof lines 3430-3431. Supplies the elements y_i making the induced map finite after tensoring with the residue field.
- Correspondence between morphisms of affine schemes and ring maps. Standard, used at proof lines 3427-3429. Supplies the translation between the geometric factorization and the ring map.

Source reference. Original lemma 00OY: sources/stacks-chow/tags.txt, source lines 912–912. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to S$ be a finite type morphism of affine schemes. Let $s \in S$. If $\dim(X_s) = d$, then there exists a factorization*

$$X \xrightarrow{\pi} \mathbf{A}^d_S \to S$$

of f such that the morphism $\pi_s : X_s \to \mathbf{A}^d_{\kappa(s)}$ of fibres over s is finite.

Proof. Write $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ and let $A \to B$ be the ring map corresponding to f. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to s. We can choose a surjection $A[x_1, \ldots, x_r] \to B$. By Algebra, Lemma 00OY there exist elements $y_1, \ldots, y_d \in A$ in the $\mathbf{Z}$-subalgebra of A generated by $x_1, \ldots, x_r$ such that the A-algebra homomorphism $A[t_1, \ldots, t_d] \to B$ sending t_i to y_i induces a finite $\kappa(\mathfrak{p})$-algebra homomorphism $\kappa(\mathfrak{p})[t_1, \ldots, t_d] \to B \otimes_A \kappa(\mathfrak{p})$. This proves the lemma. □

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